

Igor Ignashin

Research Engineer / LLM Post-Training - distillation, SFT, evaluation, multi-agent RL

Moscow, Russia | ignashin.in@phystech.edu | GitHub: [ThunderstormXX](#) | Scholar: [YXIIYbsAAAAJ](#) | Personal page: thunderstormxx.github.io/PersonalPage

PROFILE

Research engineer focused on LLM post-training and applied research. Strongest applied case: Qwen/DeepScaleR math distillation with 40.3k teacher-generated CoT traces, SFT/Forward-KL training, evaluation on five math benchmarks, and macro Pass@1 improvement from 46.97 to 54.45 against a 55.90 local RL teacher.

Worked as Research Team Lead at MIPT AI Institute / BRAIn Lab: led five small teams of 1-3 people across distillation/SFT, LLM compression, multi-agent RL/NLHF, SGD dynamics, and optimization tracks. Role: hypothesis framing, task decomposition, baseline selection, code and experiment review, and evaluation discipline.

CORE SKILLS

Languages / tools	Python, C++, SQL, Linux, Git, JSONL data workflows, YQL, DataLens, Nirvana
LLM post-training	PyTorch, Hugging Face workflows, vLLM, SFT, teacher/KL distillation, reward/grader parsing, benchmark evaluation
Research	Multi-agent RL/NLHF, DPO/PPO/GRPO framing, optimization, stochastic processes, minimax problems, SGD dynamics

EXPERIENCE

Research Team Lead / LLM Post-Training Engineer

MIPT AI Institute / BRAIn Lab / MIRAI, 2024-2026; collaboration with LAB MMO

- Led five small research teams of 1-3 people; owned research framing, baseline choice, task decomposition, code/experiment review, metric control, and reproducible artifacts.
- Built Qwen/DeepScaleR math distillation workflow: vLLM teacher generation, reward/grader parsing, cleaning, train/val split, SFT and Forward KL, benchmark evaluation, and summary tables.
- Result: 40,300 teacher traces; macro Pass@1 on AIME 2024, MATH500, AMC 2023, Minerva Math, and OlympiadBench improved from 46.97 to 54.45; local RL teacher was 55.90.
- Led multi-agent RL / NLHF / HeteroCrossPlay track: preference pairs from cross-play between heterogeneous models, verifier-reward framing, self-play comparison, and modified VERL recipe.
- Led LLM compression track around layer pruning, LoRA healing, decoder/layer importance search, attention-head merging, and quality/size trade-off evaluation.
- Led and contributed to optimization / SGD-dynamics tracks: finite-step SGD dynamics, minimax/DRO optimization, Frank-Wolfe variants, and applied optimization.

Visiting Research Student

MBZUAI, Jan 2026 - Mar 2026

- Studied LMO-based optimization methods and acceleration schemes, including formulations connected to Muon and Lookahead-type ideas.

SGD Dynamics and Multi-Agent RL Collaboration

Collaboration with Andrei Leonidov's group, 2025-present

- Worked on finite-step SGD behavior, Hessian-basis diagnostics, experiment code, figure generation, and paper-oriented result analysis.

Data Analyst Intern

Yandex, earlier role

- Automated daily B2B contractor ticket assignment/closure analytics using YQL, DataLens dashboards, and Nirvana workflows.

SELECTED PROJECTS AND ARTIFACTS

- Qwen / DeepScaleR math post-training: teacher40k and clean-v4 datasets published on Hugging Face; five-benchmark comparison board.
- HeteroCrossPlay / NLHF: multi-agent preference-learning prototype over modified VERL recipe; verifier-based reward and DPO-style update framing.
- Ridiculous-LLM-Compression: public code for layer pruning, LoRA healing, decoder search, and attention-head merging.
- SGDiffusion: reproducibility package for finite-step SGD dynamics, Langevin-style comparison, and Hessian-basis diagnostics.
- Applied optimization: wireless scheduling, traffic assignment solvers, and city logistics routing benchmarks.

SELECTED PUBLICATIONS

- Aligning Distributionally Robust Optimization with Practical Deep Learning Needs - accepted to NeurIPS 2025 workshop.
- Why SGD is not Brownian Motion: A New Perspective on Stochastic Dynamics - NeurIPS 2026 submission.
- Frank-Wolfe Modifications for Equilibrium Traffic Assignment - Computer Research and Modeling.
- Stochastic Origin Frank-Wolfe for Traffic Assignment - accepted to Journal of Mathematical Sciences, Series B.

EDUCATION

- MIPT, Applied Mathematics and Informatics: B.S. in Applied Mathematics and Physics; M.S. candidate, Department of Intelligent Data Analysis.